

Question 1

```
img = imread('Bat.jpg');  
gray_img = rgb2gray(img);  
figure, imshow(gray_img);  
title('Original Grayscale Image');
```

Original Grayscale Image



```
threshold_value = 100;  
binary_mask = imbinarize(gray_img, threshold_value/255);  
figure, imshow(binary_mask);  
title('Binary Mask');
```

Binary Mask



```
h_gaussian = fspecial('gaussian', [5,5], 2);  
img_gaussian = imfilter(gray_img, h_gaussian);  
figure, imshow(img_gaussian);  
title('Gaussian Filtered Image');
```

Gaussian Filtered Image



```
disp('Histogram for Gaussian Filtered Image:');
```

Histogram for Gaussian Filtered Image:

```
disp(histcounts(img_gaussian));
```

10853	1011795	51296	26301	17938	15555	14948	13242	110
-------	---------	-------	-------	-------	-------	-------	-------	-----

```
h_average = fspecial('average', [5,5]);  
img_avg = imfilter(gray_img, h_average);  
figure, imshow(img_avg);  
title('Average Filtered Image');
```

Average Filtered Image



```
disp('Histogram for Average Filtered Image:');
```

Histogram for Average Filtered Image:

```
disp(histcounts(img_avg));
```

9519	1008839	50084	26133	17822	15484	14868	12960	108
------	---------	-------	-------	-------	-------	-------	-------	-----

```
h_laplacian = fspecial('laplacian', 0.2);  
img_laplacian = imfilter(gray_img, h_laplacian);  
figure, imshow(img_laplacian, []);  
title('Laplacian Filtered Image');
```

Laplacian Filtered Image



```
disp('Histogram for Laplacian Filtered Image:');
```

Histogram for Laplacian Filtered Image:

```
disp(histcounts(img_laplacian));
```

4032366	0	0	0	0	456887	0	0
---------	---	---	---	---	--------	---	---

```
img_prewitt = edge(gray_img, 'prewitt');  
figure, imshow(img_prewitt);  
title('Prewitt Filtered Image');
```

Prewitt Filtered Image



```
disp('Histogram for Prewitt Filtered Image:');
```

Histogram for Prewitt Filtered Image:

```
disp(histcounts(img_prewitt));
```

5766828 234944

```
disp('Explanation and Observations:');
```

Explanation and Observations:

```
disp('1. The original image is converted to grayscale to simplify the  
analysis.');
```

1. The original image is converted to grayscale to simplify the analysis.

```
disp('2. A binary mask is created using a threshold to isolate regions of  
interest.');
```

2. A binary mask is created using a threshold to isolate regions of interest.

```
disp('3. Gaussian and Average filters are applied to reduce noise and smooth  
the image.');
```

3. Gaussian and Average filters are applied to reduce noise and smooth the image.

```
disp('4. Laplacian and Prewitt filters are used to enhance and detect edges,  
respectively.');
```

4. Laplacian and Prewitt filters are used to enhance and detect edges, respectively.

```
disp('5. The effects of these filters are visualized and can be analyzed by  
comparing the output images and histograms.');
```

5. The effects of these filters are visualized and can be analyzed by comparing the output images and histograms.

```
disp('6. Histograms provide a quantitative view of the pixel intensity  
distribution after each filtering step, offering insights into the filter  
effects.');
```

6. Histograms provide a quantitative view of the pixel intensity distribution after each filtering step, offering insights into the filter effects.